

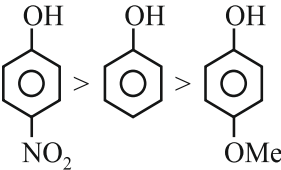
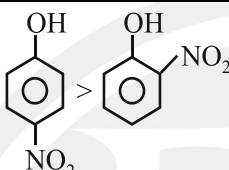
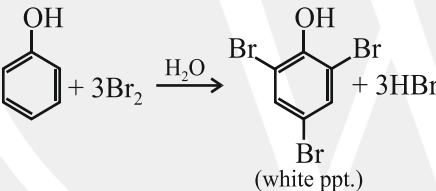
CHAPTER

5

Alcohols, Phenols and Ethers

Sr. No.	Concept	Description	Other information
1.	Name reactions:		
	(a) Dow's Process	$\text{C}_6\text{H}_5\text{X} + \text{NaOH} \xrightarrow[300 \text{ atm}]{623 \text{ K}} \text{C}_6\text{H}_5\text{O}^-\text{Na}^+ \xrightarrow{\text{HX}} \text{C}_6\text{H}_5\text{OH}$	X = Cl Reagent = NaOH, HX
	(b) Reimer Tiemann Reaction	$\text{C}_6\text{H}_5\text{OH} \xrightarrow{\text{CHX}_3 + \text{aq NaOH}} \text{C}_6\text{H}_4(\text{OH})\text{CHO} \xrightarrow{\text{H}^+} \text{Salicylaldehyde}$	X = Cl Reagent = $\text{CHX}_3 + \text{aq NaOH}$
	(c) Kolbe's Reaction	$\text{C}_6\text{H}_5\text{OH} \xrightarrow{\text{NaOH}} \text{C}_6\text{H}_5\text{ONa} \xrightarrow[\text{(ii) H}^+]{\text{(i) CO}_2} \text{C}_6\text{H}_4(\text{OH})\text{COOH}$ 2-Hydroxybenzoic acid (Salicylic acid)	Reagent = NaOH, CO_2 , H^+
	(d) Williamson synthesis	$\text{R-X} + \text{R}'-\ddot{\text{O}}\text{Na} \longrightarrow \text{R}-\ddot{\text{O}}-\text{R}' + \text{NaX}$	
	(e) Friedel-Crafts reaction		
	(i) Friedel-Crafts Alkylation	$\text{C}_6\text{H}_5\text{OH} + \text{RX} \xrightarrow[\text{CS}_2]{\text{Anhyd. AlCl}_3} \text{C}_6\text{H}_4(\text{OH})\text{R} + \text{C}_6\text{H}_4(\text{OH})\text{R}$ (Minor) (Major)	R = Alky group X = Cl, Br, I Reagent = Anhyd. AlCl_3 , CS_2
	(ii) Friedel-Crafts Acylation	$\text{C}_6\text{H}_5\text{OH} + \text{RCOX} \xrightarrow{\text{Anhyd. AlCl}_3} \text{C}_6\text{H}_4(\text{OH})\text{COR} + \text{C}_6\text{H}_4(\text{OH})\text{COR}$ (Minor) (Major)	
	(f) Oxymercuration-demercuration	$\text{R-CH=CH}_2 \xrightarrow[2. \text{NaBH}_4, \text{OH}^-]{1. \text{Hg}(\text{OAc})_2 / \text{H}_2\text{O}} \text{R-CH(OH)-CH}_3$	Markovnikov addition of water Reagent = $\text{Hg}(\text{OAc})_2 / \text{H}_2\text{O}$, NaBH_4 , OH^-

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	(g) Hydroboration-oxidation	$\text{CH}_3 - \text{CH} = \text{CH}_2 + (\text{H} - \text{BH}_2)_2 \rightarrow \text{CH}_3 - \underset{\substack{ \\ \text{H}}}{\text{CH}} - \underset{\substack{ \\ \text{BH}_2}}{\text{CH}_2}$ $\downarrow \text{CH}_3 - \text{CH} = \text{CH}_2$ $(\text{CH}_3 - \text{CH}_2 - \text{CH}_2)_3 \text{B} \xleftarrow{\text{CH}_3 - \text{CH} = \text{CH}_2} (\text{CH}_3 - \text{CH}_2 - \text{CH}_2)_2 \text{BH}$ $\text{H}_2\text{O} \downarrow 3\text{H}_2\text{O}_2, \bar{\text{O}}\text{H}$ $3\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{OH} + \text{B}(\text{OH})_3$ Propan-1-ol	Anti-Markovnikov addition of water Reagent = Diborane (BH ₃) ₂
2.	Reactivity of ROH	allyl, benzyl > 3° > 2° > 1° (Carbocation)	Reactivity ∝ stability of carbocation
3.	Acidity of alcohols	1° alcohols > 2° alcohols > 3° alcohols	Acidity is due to polar nature of OH.
4.	Ease of dehydration of alcohols	3° alcohols > 2° alcohols > 1° alcohols	
5.	Test of alcohol (Lucas test)	$\text{R} - \text{OH} + \text{HCl} \xrightarrow[\text{(Conc.)}]{\text{ZnCl}_2 / \text{Anhyd.}} \text{R} - \text{Cl} + \text{H}_2\text{O}$ Alkyl chloride	Reagent = Anhyd. ZnCl ₂ 1° – OH → No cloudiness 2° – OH → Cloudiness after 5 min 3° – OH → immediate cloudiness
6.	Physical properties of alcohol: (a) Solubility (b) Boiling point	3° (Alcohols) > 2° (alcohols) > 1° (alcohols) 1° (Alcohols) > 2° (alcohols) > 3° (alcohols)	Solubility ∝ $\frac{1}{\text{Molecular weight}}$ Solubility ∝ H-bonding B.P. ∝ $\frac{\text{Molecular weight}}{\text{Branching}}$ B.P. ∝ H-bonding
7.	S _N 1 reaction of alcohol	$\text{R} - \text{OH} \xrightarrow{\text{H}^+} \text{R} - \overset{\oplus}{\text{O}}\text{H}_2 \xrightarrow{-\text{H}_2\text{O}} \text{R}^{\oplus}$ $\xrightarrow{\text{X}^-} \text{R} - \text{X} \text{ (R may rearrange)}$	Intermediate = Carbocation
8.	S _N 2 reaction of alcohol	$\text{ROH} + \text{PCl}_5 \longrightarrow \text{RCl} + \text{POCl}_3$ $\text{ROH} + \text{PCl}_3 \longrightarrow \text{RCl} + \text{H}_3\text{PO}_3$ $\text{ROH} + \text{SOCl}_2 \xrightarrow{\text{Pyridine}} \text{RCl} + \text{SO}_2 + \text{HCl}$	Reactivity ∝ $\frac{1}{\text{Steric hindrance}}$ Reagent = Pyridine

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9.	Oxidation of alcohol (a) CrO_3 (b) PCC (c) Cu at 573 K	$1^\circ \text{OH} \xrightarrow{[\text{O}]} \text{aldehyde}$ $2^\circ \text{OH} \xrightarrow{[\text{O}]} \text{ketone}$ $3^\circ \text{OH} \xrightarrow[573\text{K}]{\text{Cu}} \text{alkene}$	$[\text{O}] = \text{CrO}_3$, PCC or Cu at 573 K
10	Acidity of phenols		Acidity $\propto$ EWG attached
11.	Boiling point of phenols		B.P. $\propto$ H-bonding B.P. $\propto$ Mwt.
12.	Test of phenols: (a) Ferric chloride test (b) Bromine-water test	$6 \text{ } \text{C}_6\text{H}_5\text{OH} + \text{FeCl}_3 \rightarrow 3\text{H}^+ + [\text{Fe}(\text{OC}_6\text{H}_5)_6]^{3-} + 3\text{HCl}$ (violet complex) 	Reagent = FeCl_3 Reagent = Br_2
13.	Boiling point of ethers	$\text{CH}_3\text{OCH}_3 > \text{C}_2\text{H}_5\text{OCH}_3 > \text{C}_2\text{H}_5\text{OC}_2\text{H}_5$	B.P. $\propto$ Mwt. B.P. of alcohol $>$ ether
14.	Cleavage of C–O bonds in ethers (a) Order of reactivity of hydrogen halides	$\text{HI} > \text{HBr} > \text{HCl}$	
15.	Grignard reagent	$\text{RCHO} + \text{R}'\text{MgX} \xrightarrow{\text{H}_2\text{O}} \text{R}-\overset{\text{R}'}{\underset{ }{\text{CH}}}-\text{OH} + \text{Mg}(\text{OH})\text{X}$	